

## 3.9 Products and Quotient Spaces

So far every new vector space in this book has been found **inside** an old one, as a subspace. This section builds new spaces in two other ways. The first glues spaces together side by side, making a product; it is the “external” direct sum promised in Chapter 1. The second, the quotient space  $V/U$ , is subtler and more important: it collapses a subspace  $U$  to a single point and keeps only what  $V$  knows beyond  $U$ . Quotients let us state precisely what a linear map remembers and what it forgets, in the three Isomorphism Theorems.

### 3.9.1 Products of vector spaces

In Chapter 1 we wrote  $V = U \oplus W$  for subspaces  $U$  and  $W$  of one space  $V$  (Definition 1.7.2). The two pieces already lived in a common space, and “direct” was a property of how they sat there. But often we want to combine two spaces that have nothing to do with each other, such as  $\mathbb{R}^2$  and  $\mathbb{R}[x]_{\leq 1}$ , into one space in which each keeps its identity. There is no common space to add inside, so we build one.

The model is  $F^2$ . A vector of  $F^2$  is a pair  $(a, b)$  of scalars, and we add and scale pairs entry by entry. Nothing in that recipe needs the entries to be scalars: the entries could be vectors, taken from any vector spaces over  $F$ .

*The product of vector spaces is the space of lists with one entry from each space, added and scaled entry by entry.*

#### Definition 3.9.1 — Product of vector spaces

Let  $V_1, \dots, V_k$  be vector spaces over the **same** field  $F$ . Their **product** (or **external direct sum**) is the Cartesian product (Definition 0.3.10)

$$V_1 \times \cdots \times V_k = \{(\mathbf{v}_1, \dots, \mathbf{v}_k) : \mathbf{v}_i \in V_i \text{ for each } i\},$$

with addition and scalar multiplication defined **entrywise**: for  $a \in F$ ,

$$(\mathbf{v}_1, \dots, \mathbf{v}_k) + (\mathbf{w}_1, \dots, \mathbf{w}_k) := (\mathbf{v}_1 + \mathbf{w}_1, \dots, \mathbf{v}_k + \mathbf{w}_k), \quad a(\mathbf{v}_1, \dots, \mathbf{v}_k) := (a\mathbf{v}_1, \dots, a\mathbf{v}_k).$$

In words: the  $i$ -th entry of a sum is the sum of the  $i$ -th entries, computed **in**  $V_i$ , and similarly for scalar multiples. Two elements are equal exactly when all their entries are equal. Every operation happens inside one  $V_i$  at a time, which is why all the  $V_i$  must share the field  $F$ : the same scalar  $a$  has to act on every entry.

**It is a vector space.** Both operations land in  $V_1 \times \cdots \times V_k$ , since  $\mathbf{v}_i + \mathbf{w}_i \in V_i$  and  $a\mathbf{v}_i \in V_i$ . Each axiom of Definition 1.1.1 is an equation between lists, so it holds when it holds in each entry, and in each entry it is the same axiom in  $V_i$ . For example, (VS6) reads  $(a + b)(\mathbf{v}_1, \dots, \mathbf{v}_k) = ((a + b)\mathbf{v}_1, \dots, (a + b)\mathbf{v}_k) = (a\mathbf{v}_1 + b\mathbf{v}_1, \dots, a\mathbf{v}_k + b\mathbf{v}_k) = a(\mathbf{v}_1, \dots, \mathbf{v}_k) + b(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , using (VS6) in each  $V_i$ . The zero vector is  $(\mathbf{0}, \dots, \mathbf{0})$ , with the zero of  $V_i$  in the  $i$ -th entry, and the additive inverse of  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is  $(-\mathbf{v}_1, \dots, -\mathbf{v}_k)$ . The other axioms are checked in exactly the same way.

**Examples.**

- **The standard example.**  $F \times \cdots \times F$  ( $n$  factors) is  $F^n$  with its usual operations.
- **Unrelated spaces.** An element of  $\mathbb{R}^2 \times \mathbb{R}[x]_{\leq 1}$  is a pair  $((a, b), c + dx)$ , and  $((1, 0), x) + ((2, 5), 3 - x) = ((3, 5), 3)$ .
- **Augmented data.** A linear system  $A\mathbf{x} = \mathbf{b}$  with  $m$  equations in  $n$  unknowns is a single element  $(A, \mathbf{b})$  of  $M_{m \times n}(F) \times F^m$ .
- **Degenerate cases.** For  $k = 1$  the product is  $V_1$  itself, with entries written in parentheses. And  $V \times \{\mathbf{0}\}$  consists of the pairs  $(\mathbf{v}, \mathbf{0})$ ; the map  $\mathbf{v} \mapsto (\mathbf{v}, \mathbf{0})$  is a linear bijection  $V \rightarrow V \times \{\mathbf{0}\}$ , so a zero factor adds nothing.

**Non-example by minimal change.** Take  $V_1 = \mathbb{C}$  as a vector space over  $\mathbb{C}$  and  $V_2 = \mathbb{R}$  over  $\mathbb{R}$ . The set  $\mathbb{C} \times \mathbb{R}$  is fine, but the entrywise rule  $i \cdot (z, t) = (iz, it)$  fails:  $it \notin \mathbb{R}$  for  $t \neq 0$ . The clause that fails is “over the **same** field”. Restricting the scalars of  $\mathbb{C}$  to  $\mathbb{R}$  (Definition 1.9.1) repairs it, and  $\mathbb{C} \times \mathbb{R}$  over  $\mathbb{R}$  is a vector space of dimension 3.

Bases of the factors combine into a basis of the product. For  $\mathbf{v} \in V_i$ , write  $\iota_i(\mathbf{v})$  for the list with  $\mathbf{v}$  in the  $i$ -th entry and  $\mathbf{0}$  in every other entry. The map  $\iota_i: V_i \rightarrow V_1 \times \cdots \times V_k$  is linear, because the operations are entrywise.

### Theorem 3.9.1 — Dimension of a Product

Let  $V_1, \dots, V_k$  be finite-dimensional vector spaces over  $F$ , and let  $(\mathbf{b}_{i1}, \dots, \mathbf{b}_{in_i})$  be a basis of  $V_i$  for each  $i$ . Then the list of all vectors  $\iota_i(\mathbf{b}_{ij})$ , for  $i = 1, \dots, k$  and  $j = 1, \dots, n_i$ , is a basis of  $V_1 \times \cdots \times V_k$ . In particular,

$$\dim(V_1 \times \cdots \times V_k) = \dim V_1 + \cdots + \dim V_k.$$

*Proof. Spanning.* Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be in the product. Since the operations are entrywise,  $(\mathbf{v}_1, \dots, \mathbf{v}_k) = \iota_1(\mathbf{v}_1) + \cdots + \iota_k(\mathbf{v}_k)$ . Each  $\mathbf{v}_i$  is a combination  $\sum_j c_{ij} \mathbf{b}_{ij}$ , and  $\iota_i$  is linear, so  $\iota_i(\mathbf{v}_i) = \sum_j c_{ij} \iota_i(\mathbf{b}_{ij})$  (Theorem 3.1.4). Hence  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is a combination of the vectors  $\iota_i(\mathbf{b}_{ij})$ .

*Independence.* Let  $\sum_{i=1}^k \sum_{j=1}^{n_i} c_{ij} \iota_i(\mathbf{b}_{ij}) = \mathbf{0}$ . The  $i$ -th entry of the left side is  $\sum_j c_{ij} \mathbf{b}_{ij}$ , because every  $\iota_{i'}(\mathbf{b}_{i'j})$  with  $i' \neq i$  has  $\mathbf{0}$  there. So  $\sum_j c_{ij} \mathbf{b}_{ij} = \mathbf{0}$  in  $V_i$  for each  $i$ , and the independence of the basis of  $V_i$  gives  $c_{ij} = 0$  for all  $j$ .

The list has  $n_1 + \cdots + n_k$  vectors, which gives the dimension (Definition 1.6.1).

■

For instance,  $\dim(\mathbb{R}^2 \times \mathbb{R}[x]_{\leq 1}) = 2 + 2 = 4$ , so this product is isomorphic to  $\mathbb{R}^4$  by Theorem 3.5.4.

How does the external product relate to the internal direct sum of Chapter 1? If  $U_1, \dots, U_k$  are subspaces of  $V$ , there is a natural linear map from the product to  $V$ : add up the entries. The sum is direct exactly when this map loses no information.

### Theorem 3.9.2 — Internal and External Direct Sums (a) Let $U_1, \dots, U_k$ be subspaces of a vector space $V$ , and let

$$S: U_1 \times \cdots \times U_k \rightarrow V, \quad S(\mathbf{u}_1, \dots, \mathbf{u}_k) = \mathbf{u}_1 + \cdots + \mathbf{u}_k.$$

Then  $S$  is linear with  $\text{im } S = U_1 + \cdots + U_k$ . It is injective if and only if the sum  $U_1 + \cdots + U_k$  is direct. In that case  $S$  is an isomorphism from  $U_1 \times \cdots \times U_k$  onto

$$U_1 \oplus \cdots \oplus U_k.$$

- (b) Conversely, let  $V_1, \dots, V_k$  be vector spaces over  $F$ . Then the images  $\iota_i(V_i)$  are subspaces of  $V_1 \times \cdots \times V_k$ , each isomorphic to  $V_i$  via  $\iota_i$ , and

$$V_1 \times \cdots \times V_k = \iota_1(V_1) \oplus \cdots \oplus \iota_k(V_k).$$

*Proof.*

- (a) For lists  $(\mathbf{u}_i)$ ,  $(\mathbf{u}'_i)$  and  $a \in F$ , regrouping the finite sums gives  $S((\mathbf{u}_i) + (\mathbf{u}'_i)) = \sum_i (\mathbf{u}_i + \mathbf{u}'_i) = S((\mathbf{u}_i)) + S((\mathbf{u}'_i))$  and  $S(a(\mathbf{u}_i)) = \sum_i a\mathbf{u}_i = aS((\mathbf{u}_i))$ , so  $S$  is linear. Its image is  $\{\mathbf{u}_1 + \cdots + \mathbf{u}_k : \mathbf{u}_i \in U_i\}$ , which is  $U_1 + \cdots + U_k$  by Definition 1.7.1. Its kernel consists of the lists with  $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{0}$ . By Theorem 3.2.3,  $S$  is injective if and only if the only such list is  $(\mathbf{0}, \dots, \mathbf{0})$ , which is condition (b) of Theorem 1.7.5, equivalent to the sum being direct. In that case  $S$ , viewed as a map onto its image, is a linear bijection, hence an isomorphism (Definition 3.5.2).
- (b) Each  $\iota_i$  is linear and injective (its kernel is  $\{\mathbf{0}\}$ , since  $\iota_i(\mathbf{v})$  has  $\mathbf{v}$  as its  $i$ -th entry), so  $\iota_i(V_i)$  is a subspace (Theorem 3.2.2) and  $\iota_i$  is a bijection onto it. Every list satisfies  $(\mathbf{v}_1, \dots, \mathbf{v}_k) = \iota_1(\mathbf{v}_1) + \cdots + \iota_k(\mathbf{v}_k)$ , so the product is the sum of the subspaces  $\iota_i(V_i)$ . If  $\iota_1(\mathbf{v}_1) + \cdots + \iota_k(\mathbf{v}_k) = \mathbf{0}$ , then  $(\mathbf{v}_1, \dots, \mathbf{v}_k) = (\mathbf{0}, \dots, \mathbf{0})$ , so every piece  $\iota_i(\mathbf{v}_i)$  is  $\mathbf{0}$ . By Theorem 1.7.5 ((b)  $\Rightarrow$  (a)) the sum is direct. ■

So internal and external direct sums are the same thing seen from two sides. An internal direct sum is a product that happens to live inside  $V$ ; a product is an internal direct sum of copies of its factors. Many books write  $\oplus$  for both. This book keeps  $\times$  for the external construction, so that  $U \oplus W$  always refers to subspaces of one space.

*Quick check.* Use Theorem 3.9.2 (a) and Rank–Nullity to explain why, for **finite-dimensional** subspaces  $U, W$  of  $V$ , the sum  $U + W$  is direct if and only if  $\dim(U + W) = \dim U + \dim W$ .

*Solution.* The addition map  $S: U \times W \rightarrow V$  has image  $U + W$ , and  $\dim(U \times W) = \dim U + \dim W$  by Theorem 3.9.1. By Theorem 3.3.1,  $\dim U + \dim W = \dim(U + W) + \dim \ker S$ . So  $\dim(U + W) = \dim U + \dim W$  exactly when  $\ker S = \{\mathbf{0}\}$ , that is, when  $S$  is injective, which by the theorem means the sum is direct. This recovers Theorem 1.7.4 (a)  $\Leftrightarrow$  (c).

## 3.9.2 Cosets

The second construction starts from a picture you have already seen. In Chapter 1 we compared the line  $U = \{x + 2y = 0\}$  through the origin with the parallel line  $L = \{x + 2y = 3\}$ , which is not a subspace. In Chapter 2 we found that the solutions of a consistent system  $A\mathbf{x} = \mathbf{b}$  form a set  $\mathbf{p} + \text{nul}(A)$ , a subspace shifted by one particular solution (Theorem 2.1.3). Both are translates of a subspace. They deserve a name.

### Definition 3.9.2 — Coset

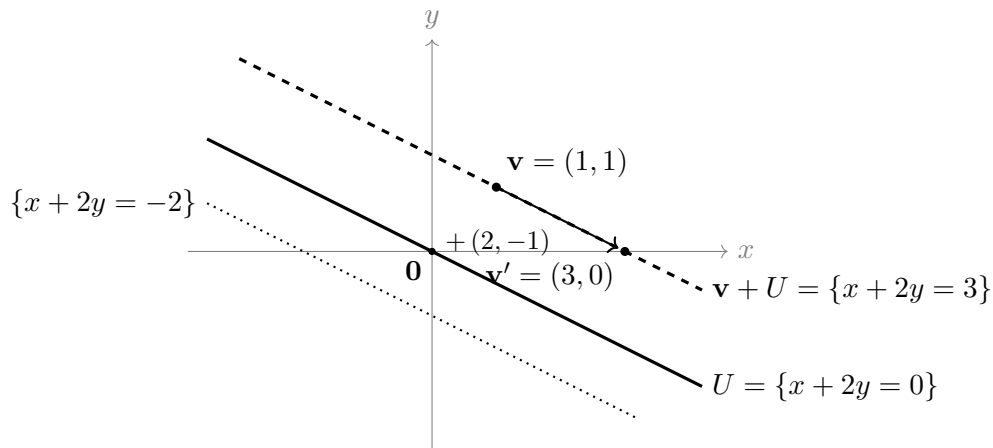
Let  $U$  be a subspace of a vector space  $V$ , and let  $\mathbf{v} \in V$ . The **coset** of  $U$  through  $\mathbf{v}$  is

the subset

$$\mathbf{v} + U := \{\mathbf{v} + \mathbf{u} : \mathbf{u} \in U\} \subseteq V.$$

Any  $\mathbf{w} \in \mathbf{v} + U$  is called a **representative** of the coset.

In  $\mathbb{R}^2$ , with  $U = \{x + 2y = 0\} = \text{span}((2, -1))$ , the coset through  $(3, 0)$  is the line  $\{x + 2y = 3\}$ : a point  $(3, 0) + t(2, -1) = (3 + 2t, -t)$  satisfies  $x + 2y = 3$ , and every solution of  $x + 2y = 3$  is of this form with  $t = -y$ . In general  $(a, b) + U = \{x + 2y = a + 2b\}$ , so the cosets of  $U$  are exactly the lines parallel to  $U$ .



The picture shows three cosets of  $U$ :  $U$  itself (solid), the line through  $\mathbf{v} = (1, 1)$  (dashed), and another parallel line (dotted). The points  $\mathbf{v} = (1, 1)$  and  $\mathbf{v}' = (3, 0)$  lie on the same dashed line, because  $\mathbf{v}' - \mathbf{v} = (2, -1) \in U$ ; so they are two representatives of **one** coset. That observation is the whole theory of cosets.

### Lemma 3.9.3 — When two cosets are equal

Let  $U$  be a subspace of  $V$ . For  $\mathbf{v}, \mathbf{w} \in V$ , write  $\mathbf{v} \sim \mathbf{w}$  if  $\mathbf{v} - \mathbf{w} \in U$ .

- (a)  $\sim$  is an equivalence relation on  $V$ , and the equivalence class of  $\mathbf{v}$  is the coset  $\mathbf{v} + U$ .
- (b)  $\mathbf{v} + U = \mathbf{w} + U$  if and only if  $\mathbf{v} - \mathbf{w} \in U$ .
- (c) Two cosets of  $U$  are either equal or disjoint, and every vector of  $V$  lies in exactly one coset.

*Proof.*

- (a) We check Definition 0.5.2. (E1)  $\mathbf{v} - \mathbf{v} = \mathbf{0} \in U$ . (E2) If  $\mathbf{v} - \mathbf{w} \in U$ , then  $\mathbf{w} - \mathbf{v} = (-1)(\mathbf{v} - \mathbf{w}) \in U$  by Theorem 1.1.7 and closure under scalar multiplication. (E3) If  $\mathbf{v} - \mathbf{w} \in U$  and  $\mathbf{w} - \mathbf{x} \in U$ , then  $\mathbf{v} - \mathbf{x} = (\mathbf{v} - \mathbf{w}) + (\mathbf{w} - \mathbf{x}) \in U$  by closure under addition. For the class,  $\mathbf{w} \sim \mathbf{v}$  means  $\mathbf{w} - \mathbf{v} = \mathbf{u}$  for some  $\mathbf{u} \in U$ , that is,  $\mathbf{w} = \mathbf{v} + \mathbf{u} \in \mathbf{v} + U$ ; so the class  $[\mathbf{v}]$  of Definition 0.5.3 is  $\mathbf{v} + U$ .
- (b) By (a) and Lemma 0.5.1 (a),  $\mathbf{v} + U = \mathbf{w} + U$  if and only if  $\mathbf{v} \sim \mathbf{w}$ , that is,  $\mathbf{v} - \mathbf{w} \in U$ .
- (c) This is Lemma 0.5.1 (b), together with  $\mathbf{v} \in \mathbf{v} + U$  (take  $\mathbf{u} = \mathbf{0}$ ).

■

### More examples.

- **Vertical lines.** In  $\mathbb{R}^3$ , let  $U = \text{span}(\mathbf{e}_3)$ , the  $z$ -axis. The coset  $(a, b, c) + U$  is the vertical line through  $(a, b, 0)$ , and  $(a, b, c) + U = (a', b', c') + U$  exactly when  $a = a'$  and  $b = b'$ . A coset is determined by the shadow of any of its points on the  $xy$ -plane.
- **Polynomials with a prescribed value.** In  $F[x]_{\leq n}$ , let  $U = \{p : p(0) = 0\}$ , a subspace (the kernel of evaluation at 0). Then  $p + U = q + U$  exactly when  $(p - q)(0) = 0$ , that is,  $p(0) = q(0)$ . So  $p + U$  is the set of all polynomials with the same constant term as  $p$ .
- **Degenerate cases.** If  $U = \{\mathbf{0}\}$ , each coset  $\mathbf{v} + U = \{\mathbf{v}\}$  is a single vector. If  $U = V$ , there is only one coset,  $\mathbf{v} + V = V$ , because  $\mathbf{v} - \mathbf{w} \in V$  always.

**Non-example by minimal change.** Try to use the line  $L = \{x + 2y = 3\}$ , which is not a subspace, in place of  $U$ . The rule “ $\mathbf{v} \sim \mathbf{w}$  if  $\mathbf{v} - \mathbf{w} \in L$ ” is not even reflexive:  $\mathbf{v} - \mathbf{v} = \mathbf{0} \notin L$ . What fails is the hypothesis that  $U$  is a subspace, which (E1), (E2) and (E3) each used.

*Warning.* A coset has many names, and it is usually not a subspace. In the example above,  $(1, 1) + U$  and  $(3, 0) + U$  are the **same** set, although  $(1, 1) \neq (3, 0)$ . Never conclude  $\mathbf{v} = \mathbf{w}$  from  $\mathbf{v} + U = \mathbf{w} + U$ ; conclude only  $\mathbf{v} - \mathbf{w} \in U$ . And  $\mathbf{v} + U$  is a subspace only when it contains  $\mathbf{0}$ , that is, when  $\mathbf{v} \in U$  and so  $\mathbf{v} + U = U$ . The dashed line above is not a subspace.

## 3.9.3 The quotient space

Now we want to treat each coset as a single point. The motivation is a kind of deliberate forgetting. Suppose we only care about vectors of  $\mathbb{R}^2$  “up to” adding something from  $U$ , for instance because we only ever measure the quantity  $x + 2y$ . Then  $(1, 1)$  and  $(3, 0)$  carry the same information, and the space of all the information we keep is the set of lines parallel to  $U$ . We would like this set to be a vector space, so that we can add and scale lines.

The recipe is the one used for  $\mathbb{Z}/n\mathbb{Z}$  in Chapter 0 (Example (Addition modulo  $n$  is well defined)): define the operations through representatives, then check that the choice of representatives does not matter.

*The quotient space  $V/U$  is  $V$  with everything in  $U$  declared to be zero.*

### Definition 3.9.3 — Quotient space

Let  $U$  be a subspace of a vector space  $V$  over  $F$ . The **quotient space**  $V/U$  (read “ $V$  modulo  $U$ ”) is the set of all cosets of  $U$ ,

$$V/U := \{\mathbf{v} + U : \mathbf{v} \in V\},$$

with the operations

$$(\mathbf{v} + U) + (\mathbf{w} + U) := (\mathbf{v} + \mathbf{w}) + U, \quad a(\mathbf{v} + U) := a\mathbf{v} + U \quad (\mathbf{v}, \mathbf{w} \in V, a \in F).$$

The **quotient map** is  $\pi: V \rightarrow V/U$ ,  $\pi(\mathbf{v}) = \mathbf{v} + U$ .

In words: an element of  $V/U$  is a whole coset, a subset of  $V$ . By [Lemma 3.9.3](#) (a),  $V/U$  is the quotient set of  $V$  by the relation  $\mathbf{v} \sim \mathbf{w} \Leftrightarrow \mathbf{v} - \mathbf{w} \in U$  (Definition 0.5.5), and  $\pi$  is its quotient map. To add two cosets, pick a representative of each, add them in  $V$ , and take the coset of the

result. To scale a coset, scale a representative. The operations are written with representatives, so before anything else we must check that they are **well defined**.

The danger is real. By Lemma 3.9.3,  $(1, 1) + U$  and  $(3, 0) + U$  are one element of  $\mathbb{R}^2/U$ . If the sum  $((1, 1) + U) + (\mathbf{w} + U)$  depended on which of the two names we used, the “definition” would assign two values to one pair of cosets, and it would not define a function at all (Definition 0.4.1). The theorem below rules this out, and then shows that the operations make  $V/U$  a vector space.

### Theorem 3.9.4 — The Quotient Space Is a Vector Space

Let  $U$  be a subspace of a vector space  $V$  over  $F$ .

- (a) The addition and scalar multiplication in Definition 3.9.3 are well defined: the right-hand sides depend only on the cosets  $\mathbf{v} + U$  and  $\mathbf{w} + U$ , not on the representatives  $\mathbf{v}$  and  $\mathbf{w}$ .
- (b) With these operations  $V/U$  is a vector space over  $F$ . Its zero vector is  $\mathbf{0} + U = U$ , and the additive inverse of  $\mathbf{v} + U$  is  $(-\mathbf{v}) + U$ .
- (c) The quotient map  $\pi: V \rightarrow V/U$  is linear and surjective, and  $\ker \pi = U$ .

*Idea.* Chapter 0’s Theorem 0.5.3 turns a function on  $V$  that is constant on classes into a function on  $V/U$ . It is stated for functions of one variable, so we use it one variable at a time. ① For scalar multiplication by a fixed  $a$ : if  $\mathbf{v} - \mathbf{v}' \in U$ , then  $a\mathbf{v} - a\mathbf{v}' = a(\mathbf{v} - \mathbf{v}') \in U$ . ② For addition, first freeze the second summand  $\mathbf{w}$  and vary the first; then show that the resulting function on cosets does not change when  $\mathbf{w}$  is replaced by an equivalent  $\mathbf{w}'$ . Each check is one line and uses exactly one closure property of  $U$ . Once the operations exist, every axiom for cosets is the same axiom in  $V$ , passed through  $\pi$ .

*Proof.* Throughout,  $\mathbf{v} \sim \mathbf{v}'$  means  $\mathbf{v} - \mathbf{v}' \in U$ , and by Lemma 3.9.3 (b),  $\mathbf{v} \sim \mathbf{v}'$  exactly when  $\mathbf{v} + U = \mathbf{v}' + U$ .

- (a) *Scalar multiplication.* Fix  $a \in F$ , and let  $g_a: V \rightarrow V/U$ ,  $g_a(\mathbf{v}) = a\mathbf{v} + U$ . If  $\mathbf{v} \sim \mathbf{v}'$ , then  $a\mathbf{v} - a\mathbf{v}' = a(\mathbf{v} - \mathbf{v}') \in U$  by (VS5) and closure of  $U$  under scalar multiplication, so  $g_a(\mathbf{v}) = g_a(\mathbf{v}')$ . By Theorem 0.5.3 (a), there is exactly one function  $\bar{g}_a: V/U \rightarrow V/U$  with  $\bar{g}_a(\mathbf{v} + U) = a\mathbf{v} + U$  for every  $\mathbf{v} \in V$ . We define  $a(\mathbf{v} + U) := \bar{g}_a(\mathbf{v} + U)$ .

*Addition, first variable.* Fix  $\mathbf{w} \in V$ , and let  $f_{\mathbf{w}}: V \rightarrow V/U$ ,  $f_{\mathbf{w}}(\mathbf{v}) = (\mathbf{v} + \mathbf{w}) + U$ . If  $\mathbf{v} \sim \mathbf{v}'$ , then  $(\mathbf{v} + \mathbf{w}) - (\mathbf{v}' + \mathbf{w}) = \mathbf{v} - \mathbf{v}' \in U$ , so  $f_{\mathbf{w}}(\mathbf{v}) = f_{\mathbf{w}}(\mathbf{v}')$ . By Theorem 0.5.3 (a), there is exactly one function  $\bar{f}_{\mathbf{w}}: V/U \rightarrow V/U$  with  $\bar{f}_{\mathbf{w}}(\mathbf{v} + U) = (\mathbf{v} + \mathbf{w}) + U$  for every  $\mathbf{v}$ .

*Addition, second variable.* Let  $\mathcal{F}$  be the set of all functions  $V/U \rightarrow V/U$ , and let  $h: V \rightarrow \mathcal{F}$ ,  $h(\mathbf{w}) = \bar{f}_{\mathbf{w}}$ . Suppose  $\mathbf{w} \sim \mathbf{w}'$ . For every  $\mathbf{v} \in V$ ,  $(\mathbf{v} + \mathbf{w}) - (\mathbf{v} + \mathbf{w}') = \mathbf{w} - \mathbf{w}' \in U$ , so

$$\bar{f}_{\mathbf{w}}(\mathbf{v} + U) = (\mathbf{v} + \mathbf{w}) + U = (\mathbf{v} + \mathbf{w}') + U = \bar{f}_{\mathbf{w}'}(\mathbf{v} + U).$$

Since every element of  $V/U$  has the form  $\mathbf{v} + U$ , the functions  $\bar{f}_{\mathbf{w}}$  and  $\bar{f}_{\mathbf{w}'}$  are equal, that is,  $h(\mathbf{w}) = h(\mathbf{w}')$ . By Theorem 0.5.3 (a) again, there is exactly one function  $\bar{h}: V/U \rightarrow \mathcal{F}$  with  $\bar{h}(\mathbf{w} + U) = \bar{f}_{\mathbf{w}}$ . We define  $X + Y := \bar{h}(Y)(X)$  for  $X, Y \in V/U$ .

Then

$$(\mathbf{v} + U) + (\mathbf{w} + U) = \bar{h}(\mathbf{w} + U)(\mathbf{v} + U) = \bar{f}_{\mathbf{w}}(\mathbf{v} + U) = (\mathbf{v} + \mathbf{w}) + U,$$

which is the formula of [Definition 3.9.3](#), and its value depends only on the two cosets.

- (b) Every element of  $V/U$  is  $\pi(\mathbf{v}) = \mathbf{v} + U$  for some  $\mathbf{v}$ , and by (a) we have  $\pi(\mathbf{v}) + \pi(\mathbf{w}) = \pi(\mathbf{v} + \mathbf{w})$  and  $a\pi(\mathbf{v}) = \pi(a\mathbf{v})$ . Let  $X = \pi(\mathbf{x})$ ,  $Y = \pi(\mathbf{y})$ ,  $Z = \pi(\mathbf{z})$  be in  $V/U$ , and  $a, b \in F$ . Each axiom follows by moving  $\pi$  outside, using the axiom in  $V$ , and moving  $\pi$  back:

- (VS1)  $X + Y = \pi(\mathbf{x} + \mathbf{y}) = \pi(\mathbf{y} + \mathbf{x}) = Y + X$ .
- (VS2)  $(X + Y) + Z = \pi((\mathbf{x} + \mathbf{y}) + \mathbf{z}) = \pi(\mathbf{x} + (\mathbf{y} + \mathbf{z})) = X + (Y + Z)$ .
- (VS3)  $X + \pi(\mathbf{0}) = \pi(\mathbf{x} + \mathbf{0}) = \pi(\mathbf{x}) = X$ , so  $\pi(\mathbf{0}) = \mathbf{0} + U = U$  is a zero vector.
- (VS4)  $X + \pi(-\mathbf{x}) = \pi(\mathbf{x} + (-\mathbf{x})) = \pi(\mathbf{0})$ , so  $\pi(-\mathbf{x}) = (-\mathbf{x}) + U$  is an additive inverse of  $X$ .
- (VS5)  $a(X + Y) = \pi(a(\mathbf{x} + \mathbf{y})) = \pi(a\mathbf{x} + a\mathbf{y}) = aX + aY$ .
- (VS6)  $(a + b)X = \pi((a + b)\mathbf{x}) = \pi(a\mathbf{x} + b\mathbf{x}) = aX + bX$ .
- (VS7)  $a(bX) = \pi(a(b\mathbf{x})) = \pi((ab)\mathbf{x}) = (ab)X$ .
- (VS8)  $1X = \pi(1\mathbf{x}) = \pi(\mathbf{x}) = X$ .

Hence  $V/U$  is a vector space over  $F$ .

- (c) The identities  $\pi(\mathbf{v} + \mathbf{w}) = \pi(\mathbf{v}) + \pi(\mathbf{w})$  and  $\pi(a\mathbf{v}) = a\pi(\mathbf{v})$  from (b) say that  $\pi$  is linear. It is surjective by the definition of  $V/U$ . Finally,  $\mathbf{v} \in \ker \pi$  means  $\mathbf{v} + U = \mathbf{0} + U$ , which by [Lemma 3.9.3](#) (b) means  $\mathbf{v} = \mathbf{v} - \mathbf{0} \in U$ . So  $\ker \pi = U$ . ■

Part (c) has a pleasant reading: **every subspace is a kernel**. Given  $U \subseteq V$ , the quotient map is a linear map whose kernel is exactly  $U$ .

### Examples.

- **Parallel lines.** In  $\mathbb{R}^2/U$  with  $U = \{x + 2y = 0\}$ , write  $\ell_c$  for the line  $\{x + 2y = c\}$ . Since  $(a, b) \in \ell_{a+2b}$ , the definition gives  $\ell_c + \ell_d = \ell_{c+d}$  and  $t\ell_c = \ell_{tc}$ : to add two parallel lines, add their “heights”  $c$  and  $d$ . The zero vector is  $\ell_0 = U$ .
- **Constant terms.** In  $F[x]_{\leq n}/U$  with  $U = \{p : p(0) = 0\}$ , the coset  $p + U$  is determined by  $p(0)$ , and  $(p + U) + (q + U) = (p + q) + U$  is determined by  $p(0) + q(0)$ . All that survives of a polynomial is its constant term.
- **Degenerate cases.**  $V/\{\mathbf{0}\}$  has the cosets  $\{\mathbf{v}\}$ , added exactly as in  $V$ : nothing is forgotten.  $V/V$  has the single element  $V$ , so it is the zero space: everything is forgotten.

**Non-example by minimal change.** Replace the line  $U$  by the union of the two axes,  $S = \text{span}(\mathbf{e}_1) \cup \text{span}(\mathbf{e}_2) \subseteq \mathbb{R}^2$ . It still contains  $\mathbf{0}$  and is still closed under scalar multiplication, but it is not closed under addition. Then “ $\mathbf{v} - \mathbf{w} \in S$ ” is not transitive:  $(1, 1) - (0, 1) = (1, 0) \in S$  and  $(0, 1) - (0, 0) = (0, 1) \in S$ , but  $(1, 1) - (0, 0) = (1, 1) \notin S$ . Concretely, the “cosets”  $(1, 1) + S$  and  $(0, 0) + S = S$  both contain  $(0, 1) = (1, 1) + (-1, 0)$ , yet they are different sets, since  $\mathbf{0} \in S$  but  $(1, 1) + \mathbf{s} = \mathbf{0}$  would need  $\mathbf{s} = (-1, -1) \notin S$ . The sets  $\mathbf{v} + S$  do not split  $\mathbb{R}^2$  into disjoint pieces, so there is no set “ $\mathbb{R}^2/S$ ” of the kind the definition needs. The clause that fails is “ $U$  is a subspace”, specifically closure under addition.

**Why this definition.** Each part of the hypothesis that  $U$  is a subspace has a job:  $\mathbf{0} \in U$  makes  $\sim$  reflexive; closure under scalar multiplication makes  $\sim$  symmetric and scaling of cosets

well defined; closure under addition makes  $\sim$  transitive and addition of cosets well defined. The non-example shows what goes wrong when one of them is missing. The notation  $V/U$  and the phrase “modulo  $U$ ” come from  $\mathbb{Z}/n\mathbb{Z}$ , where integers are identified when their difference lies in  $n\mathbb{Z}$ .

**Warning.**  $V/U$  is not a subspace of  $V$ . Its elements are subsets of  $V$ , not vectors of  $V$ , and its addition is a new operation. For example,  $\mathbb{R}^3/\text{span}(\mathbf{e}_3)$  is not the  $xy$ -plane: its elements are vertical lines. The  $xy$ -plane is a **complement** of the  $z$ -axis, one of many (Chapter 1), whereas  $V/U$  involves no choice at all. The two are isomorphic, as we show below, but they are different objects.

In finite dimension the size of a quotient is exactly what one would guess from the picture: we lose  $\dim U$  dimensions.

### Theorem 3.9.5 — Dimension of a Quotient

Let  $V$  be finite-dimensional and  $U$  a subspace of  $V$ . Then  $V/U$  is finite-dimensional and

$$\dim(V/U) = \dim V - \dim U.$$

More precisely, if  $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_s)$  is a basis of  $V$  such that  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  is a basis of  $U$ , then  $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$  is a basis of  $V/U$ .

*Proof.* By Theorem 3.9.4 (c),  $\pi: V \rightarrow V/U$  is linear, surjective and has kernel  $U$ . By the Rank–Nullity Theorem (Theorem 3.3.1),  $\text{im } \pi = V/U$  is finite-dimensional and  $\dim V = \dim U + \dim(V/U)$ , which is the formula.

For the basis, let  $\mathbf{v} + U \in V/U$  and write  $\mathbf{v} = \sum_i a_i \mathbf{u}_i + \sum_j b_j \mathbf{w}_j$ . Since  $\pi$  is linear and  $\pi(\mathbf{u}_i) = U$  is the zero vector, Theorem 3.1.4 gives  $\mathbf{v} + U = \sum_j b_j(\mathbf{w}_j + U)$ . So the list  $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$  spans  $V/U$ . Its length is  $s = \dim V - \dim U = \dim(V/U)$ , so it is a basis by Theorem 1.6.4 (b). ■

Such a basis always exists: take a basis of  $U$  and extend it (Theorem 1.6.5). So to find a basis of a quotient, **extend a basis of  $U$  to one of  $V$  and keep the cosets of the new vectors.**

#### Example. (A basis of a quotient of $\mathbb{R}^3$ )

Let  $U = \text{span}((1, 1, 1)) \subseteq \mathbb{R}^3$ . Find a basis of  $\mathbb{R}^3/U$ , and write  $(2, 5, -1) + U$  in it.

*Solution.* Extend  $(1, 1, 1)$  by  $\mathbf{e}_1$  and  $\mathbf{e}_2$ . The list  $((1, 1, 1), \mathbf{e}_1, \mathbf{e}_2)$  is independent: in  $a(1, 1, 1) + b\mathbf{e}_1 + c\mathbf{e}_2 = (a + b, a + c, a) = \mathbf{0}$ , the third entry gives  $a = 0$ , and then  $b = c = 0$ . It has length 3, so it is a basis of  $\mathbb{R}^3$  (Theorem 1.6.4). By Theorem 3.9.5,  $(\mathbf{e}_1 + U, \mathbf{e}_2 + U)$  is a basis of  $\mathbb{R}^3/U$ , which has dimension  $3 - 1 = 2$ .

To write  $(2, 5, -1) + U$ , solve  $(2, 5, -1) = a(1, 1, 1) + b\mathbf{e}_1 + c\mathbf{e}_2$ : the third entry gives  $a = -1$ , then  $b = 3$  and  $c = 6$ . Hence

$$(2, 5, -1) + U = 3(\mathbf{e}_1 + U) + 6(\mathbf{e}_2 + U),$$

and indeed  $(2, 5, -1) - (3, 6, 0) = (-1, -1, -1) \in U$ . The coordinates  $(3, 6)$  are  $(x -$

$z, y - z$  evaluated at  $(2, 5, -1)$ : they are exactly the quantities that do not change when a multiple of  $(1, 1, 1)$  is added.

The warning above said that a quotient is not a complement. The next result says how they are related: any complement is a concrete model of the quotient.

### Proposition 3.9.6 — A complement is a model of the quotient

Let  $U$  be a subspace of  $V$ , and let  $W$  be a subspace with  $V = U \oplus W$ . Then the restriction of the quotient map,  $W \rightarrow V/U$ ,  $\mathbf{w} \mapsto \mathbf{w} + U$ , is an isomorphism.

*Proof.* The restriction of the linear map  $\pi$  to  $W$  is linear. *Injective:* if  $\mathbf{w} + U = U$  with  $\mathbf{w} \in W$ , then  $\mathbf{w} \in U$  by Lemma 3.9.3 (b), so  $\mathbf{w} \in U \cap W = \{\mathbf{0}\}$  (Theorem 1.7.4); by Theorem 3.2.3 the restriction is injective. *Surjective:* given  $\mathbf{v} + U$ , write  $\mathbf{v} = \mathbf{u} + \mathbf{w}$  with  $\mathbf{u} \in U$ ,  $\mathbf{w} \in W$ ; then  $\mathbf{v} - \mathbf{w} = \mathbf{u} \in U$ , so  $\mathbf{v} + U = \mathbf{w} + U$ . A linear bijection is an isomorphism (Definition 3.5.2). ■

So every complement of  $U$  is isomorphic to  $V/U$ , which gives a second proof that all complements have the same dimension. The quotient is the choice-free object that all the complements imitate.

*Quick check.* In  $\mathbb{R}^2/U$  with  $U = \text{span}((1, 1))$ , is  $(2, 3) + U = (0, 1) + U$ ? Is  $(1, 0) + U = (0, 1) + U$ ? What is  $\dim(\mathbb{R}^2/U)$ ?

*Solution.* By Lemma 3.9.3 (b):  $(2, 3) - (0, 1) = (2, 2) \in U$ , so the first two cosets are equal.  $(1, 0) - (0, 1) = (1, -1) \notin U$ , so the second pair are different cosets. By Theorem 3.9.5,  $\dim(\mathbb{R}^2/U) = 2 - 1 = 1$ .

## 3.9.4 The First Isomorphism Theorem

A linear map  $T: V \rightarrow W$  forgets exactly the information in its kernel:  $T\mathbf{v} = T\mathbf{v}'$  if and only if  $T(\mathbf{v} - \mathbf{v}') = \mathbf{0}$ , that is,  $\mathbf{v} - \mathbf{v}' \in \ker T$ . In the language of this section,  $T$  takes the same value at two vectors exactly when they lie in the same coset of  $\ker T$ . So  $T$  is really a function of the coset, and on cosets it forgets nothing.

### Theorem 3.9.7 — First Isomorphism Theorem

Let  $T: V \rightarrow W$  be a linear map, and let  $K = \ker T$ . Then the rule

$$\bar{T}: V/K \rightarrow W, \quad \bar{T}(\mathbf{v} + K) = T\mathbf{v}$$

defines a linear map, which is injective with  $\text{im } \bar{T} = \text{im } T$ , and satisfies  $\bar{T} \circ \pi = T$ . Hence

$$V/\ker T \cong \text{im } T.$$

*Proof. Well defined.*  $K$  is a subspace by Theorem 3.2.1, so  $V/K$  is a vector space. If  $\mathbf{v} + K = \mathbf{v}' + K$ , then  $\mathbf{v} - \mathbf{v}' \in K$  by Lemma 3.9.3 (b), so  $T\mathbf{v} - T\mathbf{v}' = T(\mathbf{v} - \mathbf{v}') = \mathbf{0}$ .

Hence  $T$  is constant on each coset of  $K$ , and by Theorem 0.5.3 (a) there is exactly one function  $\bar{T}: V/K \rightarrow W$  with  $\bar{T}(\mathbf{v} + K) = T\mathbf{v}$ , that is,  $\bar{T} \circ \pi = T$ .

*Linear.* For  $\mathbf{v}, \mathbf{w} \in V$  and  $a \in F$ , by Definition 3.9.3 and the linearity of  $T$ ,

$$\bar{T}((\mathbf{v} + K) + (\mathbf{w} + K)) = \bar{T}((\mathbf{v} + \mathbf{w}) + K) = T(\mathbf{v} + \mathbf{w}) = T\mathbf{v} + T\mathbf{w} = \bar{T}(\mathbf{v} + K) + \bar{T}(\mathbf{w} + K),$$

and similarly  $\bar{T}(a(\mathbf{v} + K)) = \bar{T}(a\mathbf{v} + K) = T(a\mathbf{v}) = a\bar{T}(\mathbf{v} + K)$ .

*Injective.* If  $\bar{T}(\mathbf{v} + K) = \mathbf{0}$ , then  $T\mathbf{v} = \mathbf{0}$ , so  $\mathbf{v} \in K$  and  $\mathbf{v} + K = K$ , the zero of  $V/K$ . By Theorem 3.2.3,  $\bar{T}$  is injective.

*Image.* The values of  $\bar{T}$  are the vectors  $T\mathbf{v}$ ,  $\mathbf{v} \in V$ , so  $\text{im } \bar{T} = \text{im } T$ . Regarded as a map  $V/K \rightarrow \text{im } T$ ,  $\bar{T}$  is a linear bijection, hence an isomorphism (Definition 3.5.2). ■

So every linear map factors as “collapse the kernel, then embed”:  $T = \bar{T} \circ \pi$ , with  $\pi$  surjective and  $\bar{T}$  injective. When  $V$  is finite-dimensional, taking dimensions with Theorem 3.9.5 gives  $\dim V - \dim \ker T = \dim \text{im } T$ , which is Rank–Nullity again. The theorem is stronger than the count, because it names the isomorphism, and it holds in infinite dimension too.

The practical use runs the other way: to identify a quotient  $V/U$ , find a surjective linear map out of  $V$  whose kernel is exactly  $U$ .

### Example. (Identifying quotients)

Use the First Isomorphism Theorem to show:

- (a)  $\mathbb{R}^3/L \cong \mathbb{R}^2$ , where  $L = \text{span}((1, 1, 1))$ ;
- (b)  $\mathbb{R}[x]_{\leq n}/U \cong \mathbb{R}$ , where  $U = \{p : p(0) = 0\}$ .

*Solution.*

- (a) Let  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ ,  $T(x, y, z) = (x - z, y - z)$ , which is linear because each entry is a linear expression. It is surjective, since  $T(a, b, 0) = (a, b)$ . Its kernel is  $\{x = z, y = z\} = \{(t, t, t)\} = L$ . By Theorem 3.9.7,  $(x, y, z) + L \mapsto (x - z, y - z)$  is an isomorphism  $\mathbb{R}^3/L \rightarrow \mathbb{R}^2$ . This explains the coordinates found in Example (A basis of a quotient of  $\mathbb{R}^3$ ). The check that it is well defined is visible: adding  $(t, t, t)$  to  $(x, y, z)$  does not change  $x - z$  or  $y - z$ .
- (b) Let  $\text{ev}_0: \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}$ ,  $p \mapsto p(0)$ . It is linear by Theorem 0.8.5, surjective because the constant polynomial  $c$  maps to  $c$ , and its kernel is  $U$  by definition. By Theorem 3.9.7,  $p + U \mapsto p(0)$  is an isomorphism  $\mathbb{R}[x]_{\leq n}/U \rightarrow \mathbb{R}$ . In particular  $\dim(\mathbb{R}[x]_{\leq n}/U) = 1$  by Theorem 3.5.4, so  $\dim U = (n + 1) - 1 = n$  by Theorem 3.9.5.

### 3.9.5 The Second and Third Isomorphism Theorems

Two more identifications come from the same method: write down a surjective map, compute its kernel, and apply the First Isomorphism Theorem. The first compares a sum of subspaces with an intersection.

#### Theorem 3.9.8 — Second Isomorphism Theorem

Let  $U$  and  $W$  be subspaces of a vector space  $V$ . Then  $U \cap W$  is a subspace of  $U$ ,  $W$  is a subspace of  $U + W$ , and

$$U/(U \cap W) \rightarrow (U + W)/W, \quad \mathbf{u} + (U \cap W) \mapsto \mathbf{u} + W$$

is a well-defined isomorphism. Hence  $(U + W)/W \cong U/(U \cap W)$ .

*Idea.* Picture two planes  $U$  and  $W$  in  $\mathbb{R}^3$  meeting in a line. Modulo  $W$ , a vector of  $U + W$  is represented by its  $U$ -part alone, since the  $W$ -part is forgotten. The only vectors of  $U$  that become zero are those already in  $W$ . So the map “ $\mathbf{u} \mapsto \mathbf{u} + W$ ” from  $U$  is onto, with kernel  $U \cap W$ .

*Proof.*  $U \cap W$  is a subspace (Theorem 1.2.2) contained in  $U$ , and  $W$  is a subspace contained in  $U + W$  (Theorem 1.7.1), so both quotients exist. Let  $\pi_W: U + W \rightarrow (U + W)/W$  be the quotient map, and let  $S: U \rightarrow (U + W)/W$ ,  $S\mathbf{u} = \mathbf{u} + W$ , be its restriction to  $U$ ;  $S$  is linear because  $\pi_W$  is (Theorem 3.9.4).

*$S$  is surjective.* An element of  $(U + W)/W$  is  $(\mathbf{u} + \mathbf{w}) + W$  with  $\mathbf{u} \in U$ ,  $\mathbf{w} \in W$ . Since  $(\mathbf{u} + \mathbf{w}) - \mathbf{u} = \mathbf{w} \in W$ , Lemma 3.9.3 (b) gives  $(\mathbf{u} + \mathbf{w}) + W = \mathbf{u} + W = S\mathbf{u}$ .

$\ker S = U \cap W$ . For  $\mathbf{u} \in U$ ,  $S\mathbf{u} = W$  means  $\mathbf{u} \in W$  (Lemma 3.9.3 (b)), that is,  $\mathbf{u} \in U \cap W$ .

By Theorem 3.9.7 applied to  $S$ , the rule  $\mathbf{u} + (U \cap W) \mapsto S\mathbf{u} = \mathbf{u} + W$  is a well-defined injective linear map with image  $\text{im } S = (U + W)/W$ , hence an isomorphism. (Directly: if  $\mathbf{u} - \mathbf{u}' \in U \cap W$ , then  $\mathbf{u} - \mathbf{u}' \in W$ , so  $\mathbf{u} + W = \mathbf{u}' + W$ , and the value does not depend on the representative.)

■

If  $U$  and  $W$  are finite-dimensional, then so is  $U + W$  (it is spanned by the union of bases, Proposition 1.7.2), and Theorem 3.9.5 turns the theorem into  $\dim(U + W) - \dim W = \dim U - \dim(U \cap W)$ . That is the dimension formula for sums (Theorem 1.7.3), now as a consequence of an isomorphism rather than of a basis computation. An exercise below asks for the details.

The third theorem says that quotients can be taken in stages.

#### Theorem 3.9.9 — Third Isomorphism Theorem

Let  $U \subseteq W \subseteq V$  be subspaces. Then  $W/U = \{\mathbf{w} + U : \mathbf{w} \in W\}$  is a subspace of  $V/U$ ,

and

$$(V/U)/(W/U) \rightarrow V/W, \quad (\mathbf{v} + U) + W/U \mapsto \mathbf{v} + W$$

is a well-defined isomorphism. Hence  $(V/U)/(W/U) \cong V/W$ .

*Proof.* Let  $S: V/U \rightarrow V/W$ ,  $S(\mathbf{v} + U) = \mathbf{v} + W$ .

*Well defined.* If  $\mathbf{v} + U = \mathbf{v}' + U$ , then  $\mathbf{v} - \mathbf{v}' \in U \subseteq W$ , so  $\mathbf{v} + W = \mathbf{v}' + W$  by Lemma 3.9.3 (b). By Theorem 0.5.3 (a), applied to  $V \rightarrow V/W$ ,  $\mathbf{v} \mapsto \mathbf{v} + W$ , the rule defines a function on  $V/U$ .

*Linear and surjective.*  $S((\mathbf{v} + U) + (\mathbf{w} + U)) = S((\mathbf{v} + \mathbf{w}) + U) = (\mathbf{v} + \mathbf{w}) + W = S(\mathbf{v} + U) + S(\mathbf{w} + U)$ , and similarly for scalars, by Definition 3.9.3 in both quotients. Every  $\mathbf{v} + W$  equals  $S(\mathbf{v} + U)$ .

*Kernel.*  $S(\mathbf{v} + U) = W$  means  $\mathbf{v} \in W$ . If  $\mathbf{v} \in W$ , then  $\mathbf{v} + U \in W/U$ . Conversely, if  $\mathbf{v} + U = \mathbf{w} + U$  with  $\mathbf{w} \in W$ , then  $\mathbf{v} - \mathbf{w} \in U \subseteq W$ , so  $\mathbf{v} \in W$ . Hence  $\ker S = W/U$ . In particular  $W/U$  is a subspace of  $V/U$ , by Theorem 3.2.1.

By Theorem 3.9.7 applied to  $S$ ,  $(V/U)/(W/U) \cong \text{im } S = V/W$ , via  $(\mathbf{v} + U) + W/U \mapsto S(\mathbf{v} + U) = \mathbf{v} + W$ .

■

The notation  $W/U$  is used with two meanings, and they agree: the quotient space of  $W$  by  $U$ , and the set of cosets  $\mathbf{w} + U$  with  $\mathbf{w} \in W$ , sitting inside  $V/U$ . The coset  $\mathbf{w} + U$  is the same subset of  $V$  in both readings, and the operations agree because both are computed with representatives. For example, with  $V = \mathbb{R}^3$ ,  $U$  the  $z$ -axis and  $W$  the  $xz$ -plane,  $V/W \cong \mathbb{R}$  through the  $y$ -coordinate, and the theorem says that first forgetting  $z$  and then forgetting  $x$  leaves the same thing as forgetting both at once.

### 3.9.6 The universal property of the quotient

The proofs of the three theorems repeated one move: a linear map that vanishes on  $U$  passes to  $V/U$ . We record that move once, in its general form. It is the standard way to **define** linear maps out of a quotient.

#### Theorem 3.9.10 — Universal Property of the Quotient

Let  $U$  be a subspace of  $V$ ,  $\pi: V \rightarrow V/U$  the quotient map, and  $T: V \rightarrow W$  a linear map.

- (a) If  $U \subseteq \ker T$ , then there is **exactly one** linear map  $\bar{T}: V/U \rightarrow W$  with  $\bar{T} \circ \pi = T$ , namely  $\bar{T}(\mathbf{v} + U) = T\mathbf{v}$ . Its image is  $\text{im } T$ , and it is injective if and only if  $U = \ker T$ .
- (b) Conversely, if some function  $\bar{T}: V/U \rightarrow W$  satisfies  $\bar{T} \circ \pi = T$ , then  $U \subseteq \ker T$ .

*Proof.*

- (a) If  $\mathbf{v} - \mathbf{v}' \in U \subseteq \ker T$ , then  $T\mathbf{v} = T\mathbf{v}'$ . By Theorem 0.5.3 (a) there is exactly one

function  $\bar{T}$  with  $\bar{T}(\mathbf{v} + U) = T\mathbf{v}$ , and it is linear by the computation in the proof of [Theorem 3.9.7](#), which used only this formula. Any linear  $\bar{T}$  with  $\bar{T} \circ \pi = T$  satisfies this formula, so it is this one. The image is  $\{T\mathbf{v}\} = \text{im } T$ . The kernel of  $\bar{T}$  is  $\{\mathbf{v} + U : \mathbf{v} \in \ker T\}$ ; it is  $\{U\}$ , the zero subspace, exactly when every  $\mathbf{v} \in \ker T$  lies in  $U$ , that is, when  $\ker T \subseteq U$ , which together with  $U \subseteq \ker T$  means  $U = \ker T$ . Apply Theorem 3.2.3.

- (b) Let  $\mathbf{u} \in U$ . Then  $\pi(\mathbf{u}) = U = \pi(\mathbf{0})$ , so  $T\mathbf{u} = \bar{T}(\pi(\mathbf{u})) = \bar{T}(\pi(\mathbf{0})) = T(\mathbf{0}) = \mathbf{0}$  by Theorem 3.1.1. Hence  $U \subseteq \ker T$ .

■

In words: **linear maps out of  $V/U$  are the same as linear maps out of  $V$  that kill  $U$ .** The First Isomorphism Theorem is the case  $U = \ker T$ . For example, the map  $T: \mathbb{R}^3 \rightarrow \mathbb{R}$ ,  $T(x, y, z) = x$ , kills the  $z$ -axis  $U$ , so it induces  $\bar{T}: \mathbb{R}^3/U \rightarrow \mathbb{R}$ ,  $(x, y, z) + U \mapsto x$ . Since  $\ker T$  is the  $yz$ -plane, strictly larger than  $U$ , this  $\bar{T}$  is not injective: it also kills the coset  $\mathbf{e}_2 + U$ .

Quotients will return in Chapter 8, where an operator  $T$  with  $T(U) \subseteq U$  induces an operator on  $V/U$ , and this is the engine of induction arguments on the dimension.

## 3.9.7 Exercises

### 3.9.7.1 A. Check your understanding

#### Exercise A1.

- Define the quotient space  $V/U$  and its two operations.
- State the condition, in terms of  $\mathbf{v} - \mathbf{w}$ , for  $\mathbf{v} + U = \mathbf{w} + U$ .
- True or false:  $V/U$  is a subspace of  $V$ . Justify your answer.
- State the First Isomorphism Theorem.
- If  $U$  is a 2-dimensional subspace of  $\mathbb{R}^5$ , what is  $\dim(\mathbb{R}^5/U)$ ? What is  $\dim(\mathbb{R}^5 \times U)$ ?
- What must be checked before the formula  $(\mathbf{v} + U) + (\mathbf{w} + U) = (\mathbf{v} + \mathbf{w}) + U$  can be used as a definition, and which property of  $U$  makes the check work?

*Solution.*

- For a subspace  $U$  of  $V$ ,  $V/U = \{\mathbf{v} + U : \mathbf{v} \in V\}$ , with  $(\mathbf{v} + U) + (\mathbf{w} + U) = (\mathbf{v} + \mathbf{w}) + U$  and  $a(\mathbf{v} + U) = a\mathbf{v} + U$  ([Definition 3.9.3](#)).
- $\mathbf{v} + U = \mathbf{w} + U$  if and only if  $\mathbf{v} - \mathbf{w} \in U$  ([Lemma 3.9.3](#)).
- False. The elements of  $V/U$  are cosets, which are subsets of  $V$ , not vectors of  $V$ .
- If  $T: V \rightarrow W$  is linear, then  $\mathbf{v} + \ker T \mapsto T\mathbf{v}$  is a well-defined isomorphism  $V/\ker T \rightarrow \text{im } T$  ([Theorem 3.9.7](#)).
- $5 - 2 = 3$  by [Theorem 3.9.5](#), and  $5 + 2 = 7$  by [Theorem 3.9.1](#).
- That the right-hand side does not depend on the representatives: if  $\mathbf{v} - \mathbf{v}' \in U$  and  $\mathbf{w} - \mathbf{w}' \in U$ , then  $(\mathbf{v} + \mathbf{w}) + U = (\mathbf{v}' + \mathbf{w}') + U$ . This works because  $(\mathbf{v} + \mathbf{w}) - (\mathbf{v}' + \mathbf{w}') = (\mathbf{v} - \mathbf{v}') + (\mathbf{w} - \mathbf{w}')$  and  $U$  is closed under addition.

## 3.9.7.2 B. Practice

**Exercise B1: A basis of a quotient of  $\mathbb{R}^4$ .**

Let  $U = \text{span}((1, 0, 1, 0), (0, 1, 0, 1)) \subseteq \mathbb{R}^4$ .

- (a) Find  $\dim(\mathbb{R}^4/U)$  and a basis of  $\mathbb{R}^4/U$ .
- (b) Write  $(3, 1, 4, 1) + U$  in your basis.
- (c) Decide whether  $(1, 2, 3, 4) + U = (3, 4, 1, 2) + U$ .

*Solution.*

- (a) The two spanning vectors are independent (look at the first two entries), so  $\dim U = 2$  and  $\dim(\mathbb{R}^4/U) = 4 - 2 = 2$  by [Theorem 3.9.5](#). Extend by  $\mathbf{e}_1, \mathbf{e}_2$ . If  $a(1, 0, 1, 0) + b(0, 1, 0, 1) + c\mathbf{e}_1 + d\mathbf{e}_2 = (a + c, b + d, a, b) = \mathbf{0}$ , then  $a = b = 0$  from the last two entries and  $c = d = 0$  from the first two. So the four vectors are independent, hence a basis of  $\mathbb{R}^4$  (Theorem 1.6.4), and by [Theorem 3.9.5](#)  $(\mathbf{e}_1 + U, \mathbf{e}_2 + U)$  is a basis of  $\mathbb{R}^4/U$ .
- (b) Solve  $(3, 1, 4, 1) = (a + c, b + d, a, b)$ :  $a = 4, b = 1, c = -1, d = 0$ . Since the  $U$ -part becomes zero in the quotient,  $(3, 1, 4, 1) + U = -1(\mathbf{e}_1 + U) + 0(\mathbf{e}_2 + U)$ . Check:  $(3, 1, 4, 1) + \mathbf{e}_1 = (4, 1, 4, 1) = 4(1, 0, 1, 0) + (0, 1, 0, 1) \in U$ .
- (c) The difference is  $(1, 2, 3, 4) - (3, 4, 1, 2) = (-2, -2, 2, 2)$ . A vector  $(a, b, a, b)$  of  $U$  has equal first and third entries, but  $-2 \neq 2$ . So the difference is not in  $U$ , and the cosets are different by [Lemma 3.9.3](#).

**Exercise B2: A map that factors through a quotient.**

Let  $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$ ,  $T(p) = (p(1), p'(1))$ , and  $U = \text{span}((x - 1)^2)$ .

- (a) Show that  $U \subseteq \ker T$ , so that  $T$  induces a linear map  $\bar{T}: \mathbb{R}[x]_{\leq 2}/U \rightarrow \mathbb{R}^2$  with  $\bar{T}(p + U) = T(p)$ .
- (b) Prove that  $\bar{T}$  is an isomorphism.
- (c) Compute  $\bar{T}(x^2 + U)$ , and find the coset  $\bar{T}^{-1}(0, 1)$ .

*Solution.*

- (a)  $T$  is linear, since evaluation and differentiation are. For  $q = (x - 1)^2$ ,  $q(1) = 0$  and  $q'(1) = 2(1 - 1) = 0$ , so  $T(q) = \mathbf{0}$ , and hence  $T(cq) = \mathbf{0}$  for all  $c$ . So  $U \subseteq \ker T$ , and [Theorem 3.9.10](#) (a) gives the linear map  $\bar{T}$ .
- (b)  $T(1) = (1, 0)$  and  $T(x - 1) = (0, 1)$ , so  $T$  is surjective and  $\text{rank } T = 2$ . By Theorem 3.3.1,  $\dim \ker T = 3 - 2 = 1 = \dim U$ , so  $\ker T = U$  by Theorem 1.6.8. By [Theorem 3.9.10](#) (a),  $\bar{T}$  is injective with image  $\text{im } T = \mathbb{R}^2$ . Hence  $\bar{T}$  is a linear bijection, an isomorphism.
- (c)  $\bar{T}(x^2 + U) = T(x^2) = (1, 2)$ . Since  $T(x - 1) = (0, 1)$ ,  $\bar{T}^{-1}(0, 1) = (x - 1) + U$ , the set of all  $x - 1 + c(x - 1)^2$ .

**Exercise B3: Matrices modulo scalars.**

Let  $n \geq 1$  and  $S = \{cI_n : c \in F\} \subseteq M_n(F)$ .

- (a) Show that  $S$  is a subspace and find  $\dim(M_n(F)/S)$ .
- (b) For  $n = 2$ , find a basis of  $M_2(F)/S$ .
- (c) Show that  $A + S = B + S$  if and only if  $A - B$  is a scalar matrix, and decide whether  $\begin{pmatrix} 3 & 1 \\ 0 & 5 \end{pmatrix} + S = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} + S$ .

*Solution.*

- (a)  $S = \text{span}(I_n)$  is a subspace by Theorem 1.3.1, of dimension 1 since  $I_n \neq 0$ . By Theorem 3.9.5,  $\dim(M_n(F)/S) = n^2 - 1$ .
- (b) Extend  $(I_2)$  by  $E_{11}, E_{12}, E_{21}$ . If  $aI_2 + bE_{11} + cE_{12} + dE_{21} = \begin{pmatrix} a+b & c \\ d & a \end{pmatrix} = 0$ , then  $a = c = d = 0$  and then  $b = 0$ . So the four matrices are independent, hence a basis of  $M_2(F)$  (Theorem 1.6.4), and  $(E_{11} + S, E_{12} + S, E_{21} + S)$  is a basis of  $M_2(F)/S$  by Theorem 3.9.5.
- (c) This is Lemma 3.9.3 (b) for the subspace  $S$ . Here the difference is  $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I_2 \in S$ , so the two cosets are equal.

### 3.9.7.3 C. Going deeper

#### Exercise C1: The dimension formula from isomorphisms.

Let  $U$  and  $W$  be finite-dimensional subspaces of a vector space  $V$ .

- (a) Using Theorem 3.9.8 and Theorem 3.9.5, prove that  $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$ .
- (b) Give a second proof: apply Rank–Nullity to the addition map  $S: U \times W \rightarrow V$ ,  $S(\mathbf{u}, \mathbf{w}) = \mathbf{u} + \mathbf{w}$ , after showing that  $\mathbf{u} \mapsto (\mathbf{u}, -\mathbf{u})$  is an isomorphism from  $U \cap W$  onto  $\ker S$ .

*Solution.*

- (a)  $U + W$  is finite-dimensional, being spanned by a basis of  $U$  followed by a basis of  $W$  (Proposition 1.7.2), and  $U \cap W \subseteq U$  is finite-dimensional (Theorem 1.6.7). By Theorem 3.9.8,  $(U + W)/W \cong U/(U \cap W)$ , and isomorphic finite-dimensional spaces have equal dimension (Theorem 3.5.4). Applying Theorem 3.9.5 to both sides,

$$\dim(U + W) - \dim W = \dim U - \dim(U \cap W),$$

which rearranges to the formula.

- (b) The map  $S$  is linear with image  $U + W$  by Theorem 3.9.2 (a). Let  $J: U \cap W \rightarrow U \times W$ ,  $J\mathbf{u} = (\mathbf{u}, -\mathbf{u})$ ; it is well defined because  $\mathbf{u} \in U$  and  $-\mathbf{u} \in W$ , and linear because the operations are entrywise. It is injective, since  $(\mathbf{u}, -\mathbf{u}) = (\mathbf{0}, \mathbf{0})$  forces  $\mathbf{u} = \mathbf{0}$ . Its image lies in  $\ker S$ , since  $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ . Conversely, if  $(\mathbf{u}, \mathbf{w}) \in \ker S$ , then  $\mathbf{w} = -\mathbf{u}$ ; so  $\mathbf{u} = -\mathbf{w} \in W$ , hence  $\mathbf{u} \in U \cap W$  and  $(\mathbf{u}, \mathbf{w}) = J\mathbf{u}$ . Thus  $J$  is an isomorphism onto  $\ker S$ , and  $\dim \ker S = \dim(U \cap W)$ . By Theorem 3.9.1 and

Theorem 3.3.1,

$$\dim U + \dim W = \dim(U \times W) = \dim(U + W) + \dim(U \cap W).$$

### Exercise C2: The operator induced on a quotient.

Let  $V$  be finite-dimensional,  $T \in \mathcal{L}(V)$ , and  $U$  a subspace with  $T\mathbf{u} \in U$  for every  $\mathbf{u} \in U$ .

- (a) Prove that  $\bar{T}(\mathbf{v} + U) := T\mathbf{v} + U$  is a well-defined linear operator on  $V/U$ .
- (b) Let  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  be a basis of  $U$ , extended to a basis  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_s)$  of  $V$ . Show that

$$[T]_{\mathcal{B}} = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix},$$

where  $A \in M_m(F)$  is the matrix of the restriction  $T|_U: U \rightarrow U$  in  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ ,  $D \in M_s(F)$  is the matrix of  $\bar{T}$  in  $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$ , and  $0$  is the  $s \times m$  zero matrix.

- (c) Check (b) when  $T$  is differentiation on  $\mathbb{R}[x]_{\leq 2}$ , with  $U = \mathbb{R}[x]_{\leq 1}$  and  $\mathcal{B} = (1, x, x^2)$ .

*Hint: for (a), apply Theorem 3.9.10 to  $\pi \circ T$ .*

*Solution.*

- (a) The map  $\pi \circ T: V \rightarrow V/U$ ,  $\mathbf{v} \mapsto T\mathbf{v} + U$ , is linear as a composition of linear maps (Theorem 3.4.2). For  $\mathbf{u} \in U$ ,  $T\mathbf{u} \in U$ , so  $(\pi \circ T)(\mathbf{u}) = U$ , the zero of  $V/U$ ; thus  $U \subseteq \ker(\pi \circ T)$ . By Theorem 3.9.10 (a) there is exactly one linear map  $\bar{T}: V/U \rightarrow V/U$  with  $\bar{T}(\mathbf{v} + U) = T\mathbf{v} + U$ .
- (b) For  $j \leq m$ ,  $T\mathbf{u}_j \in U$ , so it is a combination of  $\mathbf{u}_1, \dots, \mathbf{u}_m$  only, and its  $\mathcal{B}$ -coordinates are 0 in the last  $s$  places. These first  $m$  columns, restricted to their first  $m$  entries, are the columns of the matrix  $A$  of  $T|_U$  (which maps  $U$  to  $U$  by hypothesis and is linear). For the remaining columns, write  $T\mathbf{w}_k = \sum_i b_{ik}\mathbf{u}_i + \sum_l d_{lk}\mathbf{w}_l$ . Passing to  $V/U$ , where  $\mathbf{u}_i + U = U$  is zero, gives  $\bar{T}(\mathbf{w}_k + U) = \sum_l d_{lk}(\mathbf{w}_l + U)$ . So the  $s \times s$  matrix  $(d_{lk})$  is the matrix  $D$  of  $\bar{T}$  in the basis  $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$  of Theorem 3.9.5, and the bottom-right block of  $[T]_{\mathcal{B}}$  is  $D$ .

- (c)  $T(1) = 0$ ,  $T(x) = 1$ ,  $T(x^2) = 2x$ , so  $[T]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ . Differentiation maps

$U = \mathbb{R}[x]_{\leq 1}$  into itself. The block  $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  is the matrix of  $T$  on  $U$  in  $(1, x)$ , the lower-left block is  $\begin{pmatrix} 0 & 0 \end{pmatrix}$ , and the block  $D$  is the  $1 \times 1$  matrix  $(0)$ : indeed  $\bar{T}(x^2 + U) = 2x + U = U$ , the zero coset, since  $2x \in U$ .

### Exercise C3: Counting cosets over a finite field.

Let  $F = \mathbb{F}_q$  be a finite field with  $q$  elements,  $V$  a vector space over  $F$  of dimension  $n$ , and  $U$  a subspace of dimension  $m$ .

- (a) Show that  $|V| = q^n$ .
- (b) Show that every coset  $\mathbf{v} + U$  has exactly  $|U|$  elements, and deduce  $|V/U| = q^{n-m}$  by counting.

(c) Check that (b) agrees with [Theorem 3.9.5](#).

*Solution.*

- (a) Choose a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ . By Theorem 1.5.1,  $(a_1, \dots, a_n) \mapsto a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$  is a bijection  $F^n \rightarrow V$ , and  $F^n$  has  $q^n$  elements.
- (b) The map  $U \rightarrow \mathbf{v} + U$ ,  $\mathbf{u} \mapsto \mathbf{v} + \mathbf{u}$ , is surjective by [Definition 3.9.2](#) and injective because  $\mathbf{v} + \mathbf{u} = \mathbf{v} + \mathbf{u}'$  implies  $\mathbf{u} = \mathbf{u}'$ . So  $|\mathbf{v} + U| = |U| = q^m$  by (a) applied to  $U$ . By [Lemma 3.9.3](#) (c), the cosets partition  $V$  into  $|V/U|$  disjoint sets of size  $q^m$ , so  $q^n = |V/U| \cdot q^m$ , and  $|V/U| = q^{n-m}$ .
- (c) By [Theorem 3.9.5](#),  $V/U$  is a vector space over  $\mathbb{F}_q$  of dimension  $n - m$ , so by (a) it has  $q^{n-m}$  elements, in agreement with (b). For example,  $\mathbb{F}_2^3$  modulo a line has  $8/2 = 4$  cosets.